LAPORAN PRAKTIKUM

POSTTEST 3

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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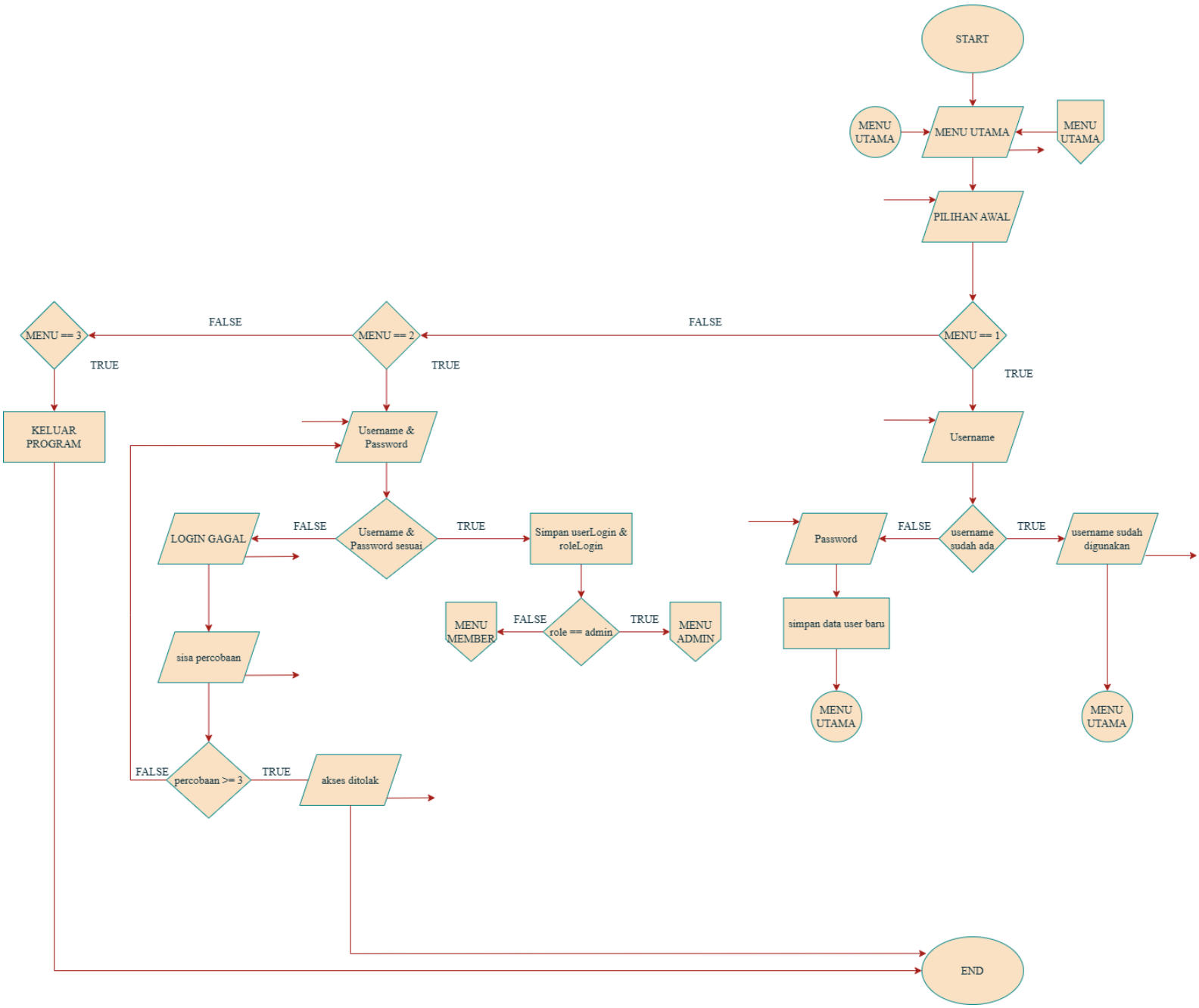
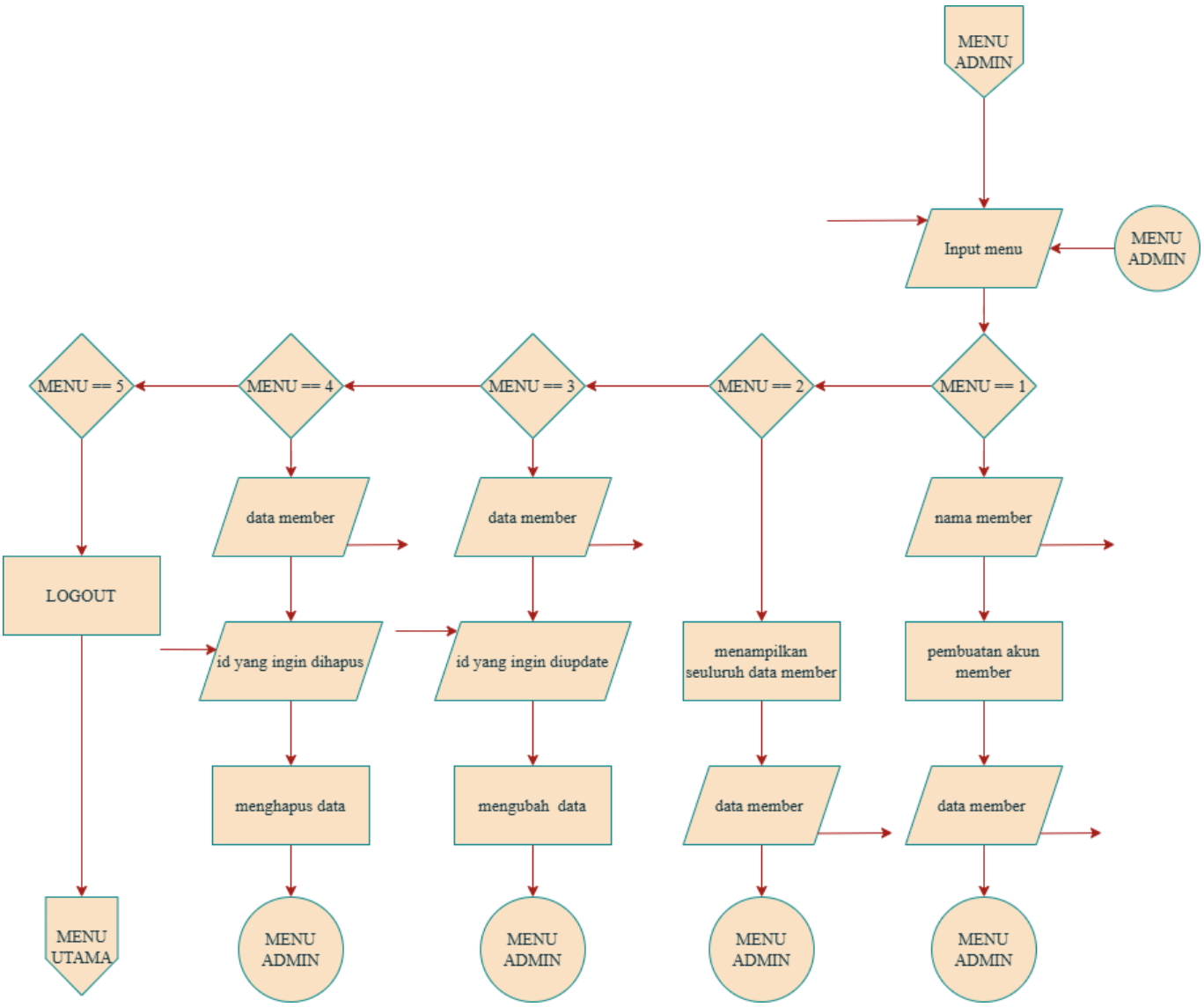
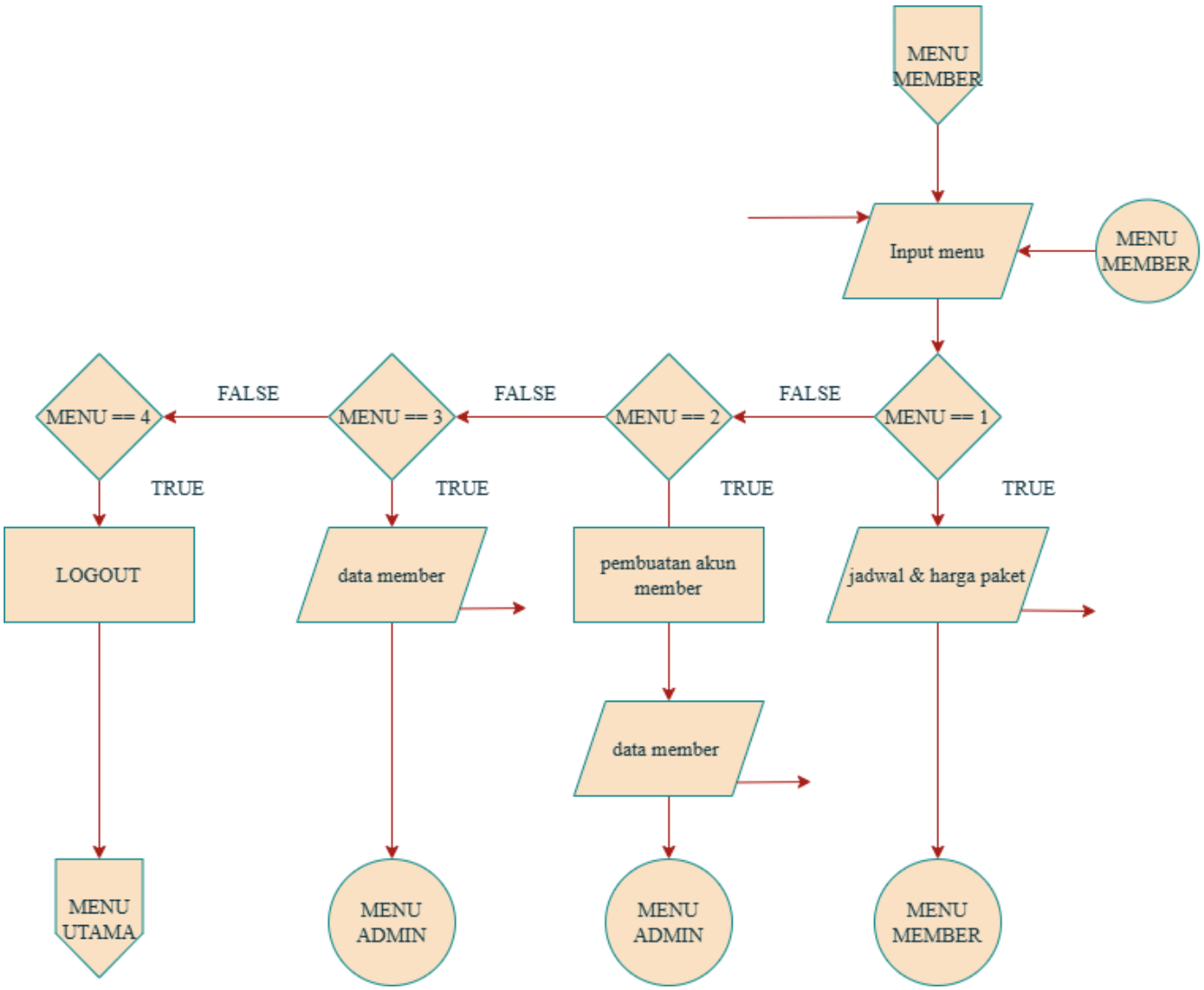
1. Flowchart


START
MENU UTAMA
MENU UTAMA
MENU UTAMA
PILIHAN AWAL
MENU == 1
FALSE
MENU == 2
FALSE
MENU == 3
TRUE
TRUE
TRUE
KELUAR PROGRAM
Username & Password
Username
LOGIN GAGAL
FALSE
Username & Password sesuai
TRUE
Simpan userLogin & roleLogin
Password
FALSE
username sudah ada
TRUE
username sudah digunakan
MENU MEMBER
FALSE
role == admin
TRUE
MENU ADMIN
simpan data user baru
sisa percobaan
MENU UTAMA
MENU UTAMA
FALSE
percobaan >= 3
TRUE
akses ditolak
END


MENU ADMIN
Input menu
MENU ADMIN
MENU == 5
MENU == 4
MENU == 3
MENU == 2
MENU == 1
data member
data member
nama member
LOGOUT
id yang ingin dihapus
id yang ingin diupdate
menampilkan seuluruh data member
pembuatan akun member
menghapus data
mengubah data
data member
data member
MENU UTAMA
MENU ADMIN
MENU ADMIN
MENU ADMIN
MENU ADMIN



1. **Flowchart Menu Utama**

- START → Program dimulai
- Tampilkan MENU UTAMA dengan 3 pilihan:
    1. Registrasi Akun Baru
    2. Login
    3. Keluar
- PILIHAN AWAL:
    1. Jika MENU == 1 (REGISTRASI):
        - Input Username
        - Cek apakah username sudah ada?

        - TRUE → Tampil "username sudah digunakan" → Kembali ke MENU UTAMA
        - FALSE → Input Password → Simpan data user baru → Kembali ke MENU UTAMA
    2. B. Jika MENU == 2 (LOGIN):
        - Input Username & Password
        - Cek kredensial (username & password sesuai?)
        - FALSE → LOGIN GAGAL → Hitung sisa percobaan
        - Jika percobaan >= 3 → AKSES DITOLAK → Program END
        - Jika percobaan < 3 → Kembali input login
        - TRUE → Simpan userLogin & roleLogin
        - Cek role == admin?
        - TRUE → Masuk ke MENU ADMIN
        - FALSE → Masuk ke MENU MEMBER
    3. Jika MENU == 3 (KELUAR):
        - KELUAR PROGRAM → END

2. Flowchart Menu Admin

- MENU == 1 → Buat Akun Member Baru
    1. Input nama member
    2. Proses pembuatan akun member
    3. Simpan data member → Kembali ke MENU ADMIN
- MENU == 2 → Tampilkan Seluruh Data Member
    1. Menampilkan semua data dalam array
    2. Tampil data member → Kembali ke MENU ADMIN
- MENU == 3 → Update Data Member
    1. Input id yang ingin diupdate
    2. Tampil data member yang akan diubah
    3. Proses mengubah data (paket, hari, jam, durasi)
    4. Simpan perubahan → Kembali ke MENU ADMIN
- MENU == 4 → Hapus Data Member
    1. Input id yang ingin dihapus
    2. Tampil data member yang akan dihapus
    3. Proses menghapus data (shift array)

4. Kembali ke MENU ADMIN

- MENU == 5 → LOGOUT → Kembali ke MENU UTAMA

3. Flowchart Menu Member

- MENU == 1 → Tampilkan jadwal & harga paket → Kembali ke MENU MEMBER
- MENU == 2 → Pembuatan akun member
    1. Input data member (nama, paket, hari, jam, durasi)
    2. Simpan data member → Kembali ke MENU MEMBER
- MENU == 3 → Tampilkan data member (data diri sendiri) → Kembali ke MENU MEMBER
- MENU == 4 → LOGOUT → Kembali ke MENU UTAMA

2. Deskripsi Singkat Program

Program ini dibuat dengan tujuan untuk memudahkan pengelolaan data member kelas padel secara terstruktur dan efisien. Program ini memungkinkan admin untuk mengelola data member (create, read, update, delete) serta member untuk mendaftar dan melihat data diri sendiri, dengan validasi paket sesi padel (SUN/MOON/STAR), pemilihan hari sesuai jadwal, dan perhitungan biaya otomatis berdasarkan durasi member.

3. Source Code

A. MENU UTAMA

```
#include <iostream>
#include <iomanip>
using namespace std;

struct user {
    string username;
    string password;
    string role;
};

struct infopaket {
    string hari;
    string jamsesi;
    int bulanMember;
    int harga;
```

```
};

struct member {
    int id;
    string nama;
    string username;
    string namapaket;
    infopaket detail;
};

#define MAX_MEMBER 100
#define MAX_user 20
#define PERTEMUAN_PER_BULAN 4

int hitungpertemuan(int bulan) {
    if (bulan <= 0) {
        return 0;
    }
    return PERTEMUAN_PER_BULAN + hitungpertemuan(bulan - 1);
}

string infohari(string namapaket) {
    if (namapaket == "SUN") {
        return "Senin-Jumat";
    } else if (namapaket == "MOON") {
        return "Senin-Jumat";
    } else if (namapaket == "STAR") {
        return "Sabtu-Minggu";
    } else {
        return "-";
    }
}

bool validasihari(string hari, string namapaket) {
    if (namapaket == "SUN" || namapaket == "MOON") {
        return (hari == "Senin" || hari == "senin" ||
                hari == "Selasa" || hari == "selasa" ||
                hari == "Rabu" || hari == "rabu" ||
                hari == "Kamis" || hari == "kamis" ||
                hari == "Jumat" || hari == "jumat");
    } else if (namapaket == "STAR") {
        return (hari == "Sabtu" || hari == "sabtu" ||
                hari == "Minggu" || hari == "minggu");
    }
    return false;
}
```

```
string inputharivalid(string namapaket) {
    string hari;
    while (true) {
        cout << "MAU PADEL DI HARI APA (" << infohari(namapaket) << "): ";
        getline(cin, hari);

        if (validasihari(hari, namapaket)) {
            if (hari.length() > 0) {
                hari[0] = toupper(hari[0]);
                for (int i = 1; i < hari.length(); i++) {
                    hari[i] = tolower(hari[i]);
                }
            }
            return hari;
        } else {
            cout << "[!] HARI TIDAK VALID! ";
            if (namapaket == "SUN" || namapaket == "MOON") {
                cout << "Paket weekday hanya bisa: Senin-Jumat" << endl;
            } else {
                cout << "Paket weekend hanya bisa: Sabtu&Minggu" << endl;
            }
        }
    }
}

int hargapaket(string namapaket, int bulan) {
    int hargaPerSesi;
    if (namapaket == "SUN") hargaPerSesi = 450;
    else if (namapaket == "MOON") hargaPerSesi = 500;
    else if (namapaket == "STAR") hargaPerSesi = 650;
    else hargaPerSesi = 0;

    int totalPertemuan = hitungpertemuan(bulan);
    return hargaPerSesi * totalPertemuan;
}

int hargapaket(string namapaket) {
    if (namapaket == "SUN") {
        return 450;
    } else if (namapaket == "MOON") {
        return 500;
    } else if (namapaket == "STAR") {
        return 650;
    } else {
        return 0;
    }
}
```

```
void infopilihanjam(string namapaket) {
    cout << "\n--- PILIH JAM SESI (max 1 pilihan) ---" << endl;

    if (namapaket == "SUN") {
        cout << "1. 06.00 - 07.00" << endl;
        cout << "2. 07.00 - 08.00" << endl;
        cout << "3. 08.00 - 09.00" << endl;
        cout << "4. 09.00 - 10.00" << endl;
        cout << "5. 10.00 - 11.00" << endl;
        cout << "6. 11.00 - 12.00" << endl;
        cout << "7. 12.00 - 13.00" << endl;
        cout << "8. 13.00 - 14.00" << endl;
        cout << "9. 14.00 - 15.00" << endl;
        cout << "10. 15.00 - 16.00" << endl;
    } else if (namapaket == "MOON" || namapaket == "STAR") {
        cout << "1. 16.00 - 17.00" << endl;
        cout << "2. 17.00 - 18.00" << endl;
        cout << "3. 18.00 - 19.00" << endl;
        cout << "4. 19.00 - 20.00" << endl;
        cout << "5. 20.00 - 21.00" << endl;
        cout << "6. 21.00 - 22.00" << endl;
        cout << "7. 22.00 - 23.00" << endl;
    }
}

string jamsesi(string namapaket, int pilihan) {
    if (namapaket == "SUN") {
        if (pilihan >= 1 && pilihan <= 10) {
            int start = 5 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        }
    } else if (namapaket == "MOON" || namapaket == "STAR") {
        if (pilihan >= 1 && pilihan <= 7) {
            int start = 15 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        }
    }
    return "-";
}

void tampilkaninfopaket() {
    cout << "\n=====================================================" << endl;
    cout << "              PAKET SESI DI MILO PADEL              " << endl;
    cout << "=====================================================" << endl;
    cout << "1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi" << endl;
    cout << "2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi" << endl;
```

```
    cout << "3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi" << endl;
    cout << "=========================================================" << endl;
    cout << "[FYI] 1 bulan = 4 pertemuan (1x seminggu)" << endl;
    cout << "=========================================================" << endl;
}

void buatmember(member members[], int &jumlahmember, string username) {
    for (int i = 0; i < jumlahmember; i++) {
        if (members[i].username == username) {
            cout << "\n[!] MEMBER SUDAH ADA" << endl;
            cout << "    menu 'lihat data saya' buat lihat data sendiri" << endl;
            return;
        }
    }

    if (jumlahmember < MAX_MEMBER) {
        cout << "\n--- BUAT AKUN MEMBER ---" << endl;
        member baru;
        baru.id = jumlahmember + 1;
        baru.username = username;

        cout << "Nama: ";
        getline(cin, baru.nama);

        tampilkaninfopaket();
        cout << "Paket (SUN/MOON/STAR): ";
        getline(cin, baru.namapaket);

        if (baru.namapaket != "SUN" && baru.namapaket != "MOON" && baru.namapaket !=
"STAR") {
            cout << "[!] MAAF, PAKET INI GA ADA!" << endl;
            return;
        }

        baru.detail.hari = inputharivalid(baru.namapaket);

        infopilihanjam(baru.namapaket);
        cout << "JAM BERAPA (1-" << (baru.namapaket == "SUN" ? "10" : "7") << "): ";
        int pilihanJam;
        cin >> pilihanJam;
        cin.ignore();
        baru.detail.jamsesi = jamsesi(baru.namapaket, pilihanJam);

        cout << "\nMAU JOIN BERAPA BULAN: ";
        cin >> baru.detail.bulanMember;
        cin.ignore();
```

```
        baru.detail.harga = hargapaket(baru.namapaket);

        members[jumlahmember] = baru;
        jumlahmember++;

        int totalPertemuan = hitungpertemuan(baru.detail.bulanMember);
        int total = hargapaket(baru.namapaket, baru.detail.bulanMember);

        cout << "\n[+] YEY AKUN MEMBER UDAH ADA!" << endl;
        cout << "     Paket          : " << baru.namapaket << endl;
        cout << "     Hari           : " << baru.detail.hari << endl;
        cout << "     Jam sesi       : " << baru.detail.jamsesi << endl;
        cout << "     Durasi         : " << baru.detail.bulanMember << " bulan" << endl;
        cout << "     Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
        cout << "     Total Pertemuan : " << totalPertemuan << "x" << endl;
        cout << "     Harga /Sesi     : Rp " << baru.detail.harga << ".000" << endl;
        cout << "     TOTAL BAYAR   : Rp " << total << ".000 " << endl;

    } else {
        cout << "\n[!] KAPASITAS MEMBER UDAH FULL!" << endl;
    }
}

void bacasemuadata(member members[], int jumlahmember) {
    if (jumlahmember == 0) {
        cout << "\n[!] DATA MEMBER MASIH KOSONG" << endl;
    } else {
        cout <<
"\n==========================================================================
================" << endl;
        cout << "                              DAFTAR MEMBER PADEL                              " << endl;
        cout <<
"==========================================================================
================" << endl;
        cout << left << setw(4) << "ID"
             << setw(15) << "Nama member"
             << setw(15) << "username"
             << setw(8)  << "Paket"
             << setw(10) << "Hari"
             << setw(12) << "Jam sesi"
             << setw(10) << "Bulan"
             << setw(12) << "Total" << endl;
        cout << "--------------------------------------------------------------------------" << endl;

        for (int i = 0; i < jumlahmember; i++) {
            int totalPertemuan = hitungpertemuan(members[i].detail.bulanMember);
            int total = hargapaket(members[i].namapaket, members[i].detail.bulanMember);
```

```
        cout << left << setw(4) << members[i].id
            << setw(18) << members[i].nama
            << setw(15) << members[i].username
            << setw(8) << members[i].namapaket
            << setw(12) << members[i].detail.hari
            << setw(12) << members[i].detail.jamsesi
            << setw(10) << (to_string(members[i].detail.bulanMember) + " bln")
            << setw(12) << ("Rp " + to_string(total) + "K") << endl;
    }
    cout <<
"==================================================================
==========" << endl;
  }
}

void bacadatasaya(member members[], int jumlahmember, string username) {
  bool ditemukan = false;

  for (int i = 0; i < jumlahmember; i++) {
    if (members[i].username == username) {
      int totalPertemuan = hitungpertemuan(members[i].detail.bulanMember);
      int total = hargapaket(members[i].namapaket, members[i].detail.bulanMember);

      cout << "\n=====================================" << endl;
      cout << "            DATA MEMBER ANDA          " << endl;
      cout << "=====================================" << endl;
      cout << "ID              : " << members[i].id << endl;
      cout << "Nama            : " << members[i].nama << endl;
      cout << "Paket           : " << members[i].namapaket << endl;
      cout << "Hari            : " << members[i].detail.hari << endl;
      cout << "Jam sesi        : " << members[i].detail.jamsesi << endl;
      cout << "Durasi          : " << members[i].detail.bulanMember << " bulan" << endl;
      cout << "Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
      cout << "Total Pertemuan : " << totalPertemuan << "x" << endl;
      cout << "Harga /Sesi     : Rp " << members[i].detail.harga << ".000" << endl;
      cout << " TOTAL BAYAR   : Rp " << total << ".000 " << endl;
      cout << "=====================================" << endl;
      ditemukan = true;
      break;
    }
  }

  if (!ditemukan) {
    cout << "\n[!] BELUM JOIN MEMBER!" << endl;
    cout << "    YUK buat akun dulu" << endl;
  }
```

```
}

void updatedatamember(member members[], int jumlahmember) {
    if (jumlahmember == 0) {
        cout << "\n[!] DATA MEMBER MASIH KOSONG" << endl;
    } else {
        cout << "\nDaftar Member:" << endl;
        for (int i = 0; i < jumlahmember; i++) {
            cout << "  [" << members[i].id << "] " << members[i].nama
                << " | Paket: " << members[i].namapaket << endl;
        }

        int index;
        cout << "\nID yang mau diupdate: ";
        cin >> index;
        cin.ignore();

        if (index > 0 && index <= jumlahmember) {
            int idx = index - 1;
            cout << "\n--- UPDATE DATA MEMBER ---" << endl;
            cout << "Nama: " << members[idx].nama << endl;
            cout << "Paket sekarang: " << members[idx].namapaket << endl;

            tampilkaninfopaket();
            cout << "Paket Baru (SUN/MOON/STAR): ";
            getline(cin, members[idx].namapaket);

            members[idx].detail.hari = inputharivalid(members[idx].namapaket);

            infopilihanjam(members[idx].namapaket);
            cout << "Pilih jam sesi (1-" << (members[idx].namapaket == "SUN" ? "10" : "7") << "): ";
            int pilihanJam;
            cin >> pilihanJam;
            cin.ignore();
            members[idx].detail.jamsesi = jamsesi(members[idx].namapaket, pilihanJam);

            cout << "MAU JOIN BERAPA BULAN: ";
            cin >> members[idx].detail.bulanMember;
            cin.ignore();

            members[idx].detail.harga = hargapaket(members[idx].namapaket);

            int totalPertemuan = hitungpertemuan(members[idx].detail.bulanMember);
            int total = hargapaket(members[idx].namapaket, members[idx].detail.bulanMember);

            cout << "\n[+] Paket member \"" << members[idx].nama << "\" YEY UPDATE
BERHASIL!" << endl;
```

```
            cout << "    Total Pertemuan : " << totalPertemuan << "x" << endl;
            cout << "     TOTAL BAYAR   : Rp " << total << ".000 " << endl;
        } else {
            cout << "\n[!] ID MEMBER GA ADA!" << endl;
        }
    }
}

void deletedatamember(member members[], int &jumlahmember) {
    if (jumlahmember == 0) {
        cout << "\n[!] DATA MEMBER MASIH KOSONG!" << endl;
    } else {
        cout << "\nDaftar Member:" << endl;
        for (int i = 0; i < jumlahmember; i++) {
            cout << "  [" << members[i].id << "] " << members[i].nama << endl;
        }

        int index;
        cout << "\nID yang mau dihapus: ";
        cin >> index;
        cin.ignore();

        if (index > 0 && index <= jumlahmember) {
            for (int i = index - 1; i < jumlahmember - 1; i++) {
                members[i] = members[i + 1];
            }
            jumlahmember--;
            cout << "\n[+] MEMBER ID INI " << index << " BERHASIL DIHAPUS!" << endl;
        } else {
            cout << "\n[!] ID MEMBER GA ADA!" << endl;
        }
    }
}
```

B. MENU ADMIN

```
int main() {
    user users[MAX_user] = {
        {"risma", "102", "member"},
        {"admin", "milopadel", "admin"}
    };
    int jumlahuser = 2;
```

```
member members[MAX_MEMBER];
int jumlahmember = 0;

string inputnama, inputpw;
string userLogin, roleLogin;

while (true) {
    cout << "!=================================!" << endl;
    cout << "        WELCOME TO MILO PADEL        " << endl;
    cout << "!=================================!" << endl;

    cout << "\n--- MENU UTAMA---" << endl;
    cout << "1. Registrasi Akun Baru" << endl;
    cout << "2. Login" << endl;
    cout << "3. Keluar" << endl;
    cout << "Pilihan: ";
    int pilihanawal;
    cin >> pilihanawal;
    cin.ignore();

    if (pilihanawal == 1) {
        cout << "\n--- REGISTRASI AKUN ---" << endl;
        cout << "username: ";
        getline(cin, inputnama);

        bool usernameada = false;
        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usernameada = true;
                break;
            }
        }

        if (usernameada) {
            cout << "[!] username \"" << inputnama << "\" SUDAH DIGUNAKAN!" << endl;
            continue;
        }

        cout << "Password: ";
        getline(cin, inputpw);
        users[jumlahuser].username = inputnama;
        users[jumlahuser].password = inputpw;
        users[jumlahuser].role = "user";
        jumlahuser++;

        cout << "\n[+] REGIS BERHASIL! YUK LOGIN" << endl;
        cout << endl;
```

```
    }

    else if (pilihanawal == 2) {
        cout << "\n--- LOGIN user ---" << endl;

        bool loginSukses = false;

        for (int percobaan = 0; percobaan < 3; percobaan++) {
            cout << "username: ";
            getline(cin, inputnama);
            cout << "Password: ";
            getline(cin, inputpw);

            bool usnbenar = false;
            bool pwbenar = false;

            for (int i = 0; i < jumlahuser; i++) {
                if (inputnama == users[i].username) {
                    usnbenar = true;
                    if (inputpw == users[i].password) {
                        pwbenar = true;
                        userLogin = inputnama;
                        roleLogin = users[i].role;
                        break;
                    }
                }
            }

            if (usnbenar && pwbenar) {
                loginSukses = true;
                break;
            }

            cout << "\n[!] LOGIN GAGAL" << endl;
            if (!usnbenar) {
                cout << "    -> userNAME GA ADA NIH!" << endl;
            } else {
                cout << "    -> PASSWORD SALAH!" << endl;
            }

            if (percobaan == 2) {
                cout << "\n[!] AKSES DITOLAK! GAGAL LOGIN 3x." << endl;
                return 0;
            }

            cout << "    -> Sisa percobaan: " << (2 - percobaan) << endl << endl;
        }
```

```
if (!loginSukses) {
    return 0;
}

int pilihan;
bool menuberjalan = true;

do {
    cout << endl;
    cout << "=============================================" << endl;
    cout << "         login sebagai: " << userLogin << " (" << roleLogin << ")" << endl;
    cout << "         mau ngapain nih???              " << endl;
    cout << "=============================================" << endl;

    if (roleLogin == "admin") {
        cout << "1. Buat Akun Member Baru      " << endl;
        cout << "2. Tampilkan Semua Data       " << endl;
        cout << "3. Update Paket Member    " << endl;
        cout << "4. Hapus Member" << endl;
        cout << "5. Logout                     " << endl;
        cout << "=========================================" << endl;
        cout << "Pilih menu (1-5): ";
        cin >> pilihan;
        cin.ignore();

        if (pilihan < 1 || pilihan > 5) {
            cout << "[!] PILIHAN TIDAK VALID! hanya 1-5." << endl;
            continue;
        }

        switch (pilihan) {
            case 1:
                buatmember(members, jumlahmember, "admin_managed");
                break;
            case 2:
                bacasemuadata(members, jumlahmember);
                break;
            case 3:
                updatedatamember(members, jumlahmember);
                break;
            case 4:
                deletedatamember(members, jumlahmember);
                break;
            case 5:
                cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
                menuberjalan = false;
```

```
                break;
            }
```

C. MENU MEMBER

```
        } else {
            cout << "1. Lihat Jadwal & Harga Paket   " << endl;
            cout << "2. Buat Akun Member        " << endl;
            cout << "3. Lihat Data Saya             " << endl;
            cout << "4. Logout                    " << endl;
            cout << "==========================================" << endl;
            cout << "Pilih menu (1-4): ";
            cin >> pilihan;
            cin.ignore();

            if (pilihan < 1 || pilihan > 4) {
                cout << "[!] PILIHAN TIDAK VALID! hanya 1-4." << endl;
                continue;
            }

            switch (pilihan) {
                case 1:
                    tampilkaninfopaket();
                    break;
                case 2:
                    buatmember(members, jumlahmember, userLogin);
                    break;
                case 3:
                    bacadatasaya(members, jumlahmember, userLogin);
                    break;
                case 4:
                    cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
                    menuberjalan = false;
                    break;
            }
        }

    } while (menuberjalan);

} else if (pilihanawal == 3) {
    cout << "\nTHANK YOU! see you milovers." << endl;
    break;
} else {
    cout << "[!] PILIHAN TIDAK VALID!" << endl;
}
```

```
    }

    return 0;
}
```

4. Hasil Output

```
!=================================!
        WELCOME TO MILO PADEL
!=================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: risma
[!] Username "risma" SUDAH DIGUNAKAN!
!=================================!
        WELCOME TO MILO PADEL
!=================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.1 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: dapa
Password: 111

[+] REGIS BERHASIL! YUK LOGIN

!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.2 Hasil Output

```
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

============================================
          login sebagai: risma (member)
          mau ngapain nih???
============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
============================================
Pilih menu (1-4):
```

Gambar 4.3 Hasil Output

```
--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: dapa
Password: 123

[!] LOGIN GAGAL
    -> PASSWORD SALAH!
    -> Sisa percobaan: 2

Username: 124
Password: dapa

[!] LOGIN GAGAL
    -> USERNAME GA ADA NIH!
    -> Sisa percobaan: 1

Username: dapa
Password: 124

[!] LOGIN GAGAL
    -> PASSWORD SALAH!

[!] AKSES DITOLAK! GAGAL LOGIN 3x.
PS C:\Users\lenov\Downloads\praktikum_apl\post-tes
```

Gambar 4.4 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 1

=====================================================
               PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
```

Gambar 4.5 Hasil Output

```
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): sun
[!] MAAF, PAKET INI GA ADA!
```

Gambar 4.6 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): SUN
MAU PADEL DI HARI APA (Senin-Jumat): sabtu
[!] HARI TIDAK VALID! Paket weekday hanya bisa: Senin-Jumat
MAU PADEL DI HARI APA (Senin-Jumat): kamis
```

Gambar 4.7 Hasil Output

```
--- PILIH JAM SESI (max 1 pilihan) ---
1. 06.00 - 07.00
2. 07.00 - 08.00
3. 08.00 - 09.00
4. 09.00 - 10.00
5. 10.00 - 11.00
6. 11.00 - 12.00
7. 12.00 - 13.00
8. 13.00 - 14.00
9. 14.00 - 15.00
10. 15.00 - 16.00
JAM BERAPA (1-10): 11

MAU JOIN BERAPA BULAN: 2

[+] YEY AKUN MEMBER UDAH ADA!
      Paket           : SUN
      Hari            : Kamis
      Jam sesi        : -
      Durasi          : 2 bulan
      Pertemuan/Bulan : 4x
      Total Pertemuan : 8x
      Harga /Sesi     : Rp 450.000
      TOTAL BAYAR   : Rp 3600.000

==========================================
          login sebagai: risma (member)
          mau ngapain nih???
==========================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
==========================================
Pilih menu (1-4):
```

Gambar 4.8 Hasil Output

```
Pilih menu (1-4): 3

=====================================
              DATA MEMBER ANDA
========================================
ID               : 1
Nama             : risma
Paket            : SUN
Hari             : Kamis
Jam sesi         : -
Durasi           : 2 bulan
Pertemuan/Bulan  : 4x
Total Pertemuan  : 8x
Harga /Sesi      : Rp 450.000
 TOTAL BAYAR    : Rp 3600.000
========================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
```

Gambar 4.9 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 3

THANK YOU! see you milovers.
PS C:\Users\lenov\Downloads\praktikum_apl\post-t
```

Gambar 4.10 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: admin
Password: milopadel

=============================================
          login sebagai: admin (admin)
          mau ngapain nih???
=============================================
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Logout
========================================
Pilih menu (1-5):
```

Gambar 4.11 Hasil Output

5. Langkah-langkah GIT

5.1 GIT Add

Git Add adalah perintah yang digunakan untuk menambahkan semua file yang ada di folder tersebut ke staging area. Titik (.) berarti semua file yang berubah atau baru dibuat akan disiapkan untuk disimpan.

```
PS C:\Users\lenov\Downloads\praktikum_apl> git add .
```

Gambar 4.12 Git Add

5.2 GIT Commit

Git Commit adalah Git Commit adalah menyimpan perubahan dari staging area ke local repository. -m "finish pt 3" adalah pesan commit yang menjelaskan perubahan ini untuk "finish pt 3" (pertemuan 3).

- 7 files changed → Ada 7 file yang diubah/ditambahkan
- 720 insertions(+) → Sekitar 720 baris kode baru ditambahkan

- 1 deletion(-) → 1 baris kode dihapus

Detail perubahan:

1. rename kelas/{pertemuan-1 => pertemuan-2}/pertemuan-2.cpp (100%) → File pertemuan-2.cpp dipindahkan dari folder pertemuan-1 ke pertemuan-2rename kelas/{pertemuan-1 => pertemuan-2}/pertemuan-2.exe (99%) → File pertemuan-2.exe dipindahkan dari folder pertemuan-1 ke pertemuan-2
2. create mode 100644 kelas/pertemuan-3/pertemuan-3.cpp → File pertemuan-3.cpp baru dibuat
3. create mode 100644 kelas/pertemuan-3/pertemuan-3.exe → File pertemuan-3.exe baru dibuat
4. create mode 100644 post-test/post-test-apl-3/2509106102-RISMAYANTY-PT-3.cpp → File post-test PT-3.cpp baru dibuat
5. create mode 100644 post-test/post-test-apl-3/2509106102-RISMAYANTY-PT-3.exe → File post-test PT-3.exe baru dibuat

```
PS C:\Users\lenov\Downloads\praktikum_apl> git commit -m "finish pt 3"
[main fcb0aca] finish pt 3
 7 files changed, 720 insertions(+), 1 deletion(-)
 rename kelas/{pertemuan-1 => pertemuan-2}/pertemuan-2.cpp (100%)
 rename kelas/{pertemuan-1 => pertemuan-2}/pertemuan-2.exe (99%)
 create mode 100644 kelas/pertemuan-3/pertemuan-3.cpp
 create mode 100644 kelas/pertemuan-3/pertemuan-3.exe
 create mode 100644 post-test/post-test-apl-3/2509106102-RISMAYANTY-PT-3.cpp
 create mode 100644 post-test/post-test-apl-3/2509106102-RISMAYANTY-PT-3.exe
```

Gambar 4.13 Git Commit

5.3 GIT Push

Git Push adalah mengunggah commit dari local repository ke remote repository (GitHub).

- Enumerating objects: 21, done → Git menghitung 21 objek yang akan dikirim
- Counting objects: 100% (21/21), done → Semua objek berhasil dihitung
- Delta compression using up to 16 threads → Git menggunakan kompresi delta dengan 16 thread untuk efisiensi
- Compressing objects: 100% (15/15), done → 15 objek berhasil dikompresi
- Writing objects: 100% (15/15), 1.19 MiB | 486.00 KiB/s, done → 15 objek (ukuran 1.19 MB) berhasil dikirim dengan kecepatan 486 KB/s
- Total 15 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0) → Total 15 objek dengan 5 delta (perbedaan), tidak ada objek yang digunakan ulang
- remote: Resolving deltas: 100% (5/5), completed with 3 local objects → Server GitHub berhasil memproses 5 delta
- remote: This repository moved. Please use the new location: → Repository dipindahkan ke lokasi baru

- To https://github.com/rismayanty/praktikum-apl.git → Berhasil push ke repository GitHub
- b493976..fcb0aca main -> main → Commit hash dari b493976 ke fcb0aca di branch main
- branch 'main' set up to track 'origin/main' → Branch lokal main sekarang terhubung dengan branch main di remote

```
PS C:\Users\lenov\Downloads\praktikum_apl> git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 16 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (15/15), 1.19 MiB | 486.00 KiB/s, done.
Total 15 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 3 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/rismayanty/praktikum-apl.git
To https://github.com/rismayanty/praktikum.apl.git
   b493976..fcb0aca  main -> main
branch 'main' set up to track 'origin/main'.
```

Gambar 4.14 Git Push